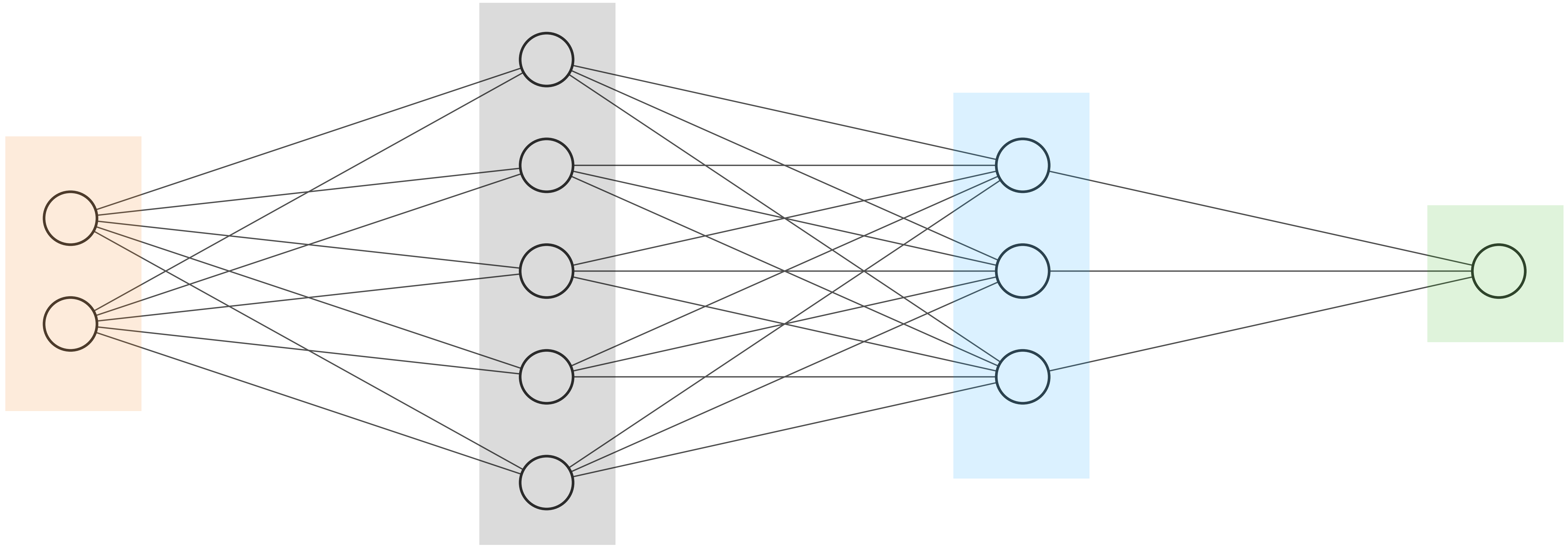
Let's see how many parameters are there in the Neural Network below.



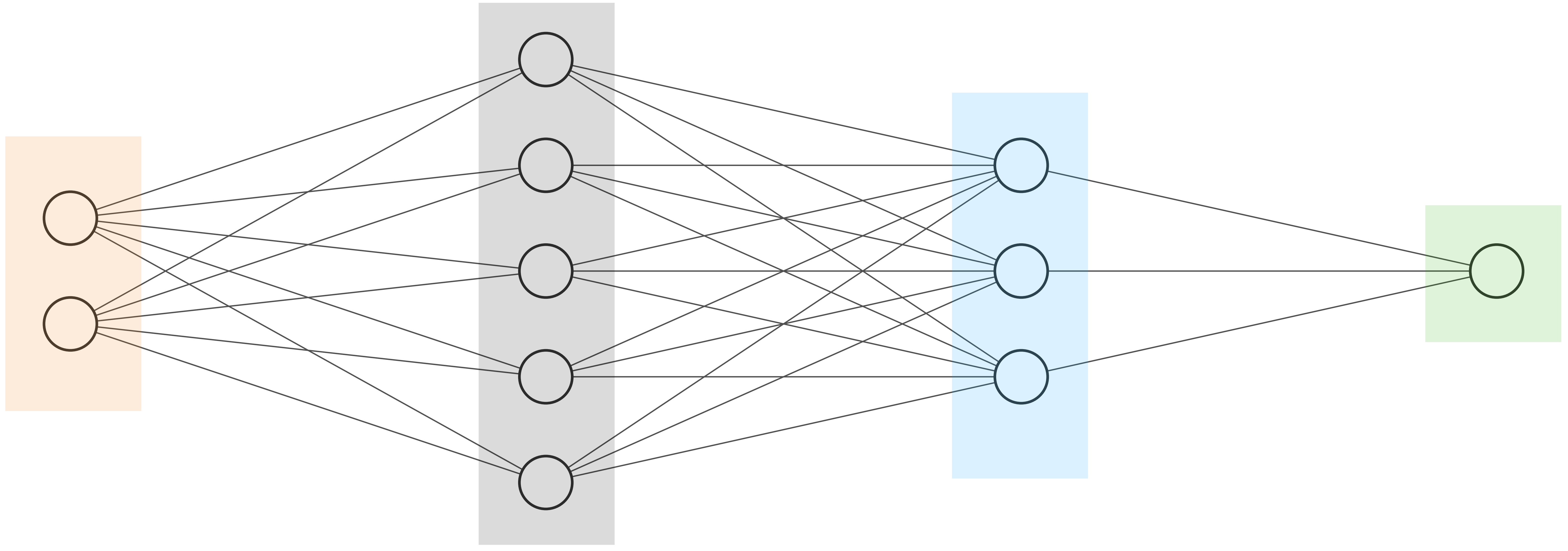
Input Layer $\in \mathbb{R}^2$ Hidden Layer $\in \mathbb{R}^5$ Hidden Layer $\in \mathbb{R}^3$ Output Layer $\in \mathbb{R}^1$

$$\begin{Bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{14} \\ h_{15} \end{Bmatrix} = \begin{bmatrix} W^{(1)}_{11} & W^{(1)}_{12} \\ W^{(1)}_{21} & W^{(1)}_{22} \\ W^{(1)}_{31} & W^{(1)}_{32} \\ W^{(1)}_{41} & W^{(1)}_{42} \\ W^{(1)}_{51} & W^{(1)}_{52} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} + \begin{Bmatrix} b_{11} \\ b_{12} \\ b_{13} \\ b_{14} \\ b_{15} \end{Bmatrix}$$

$$\begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix} = \begin{bmatrix} W^{(2)}_{11} & W^{(2)}_{12} & W^{(2)}_{13} & W^{(2)}_{14} & W^{(2)}_{15} \\ W^{(2)}_{21} & W^{(2)}_{22} & W^{(2)}_{23} & W^{(2)}_{24} & W^{(2)}_{25} \\ W^{(2)}_{31} & W^{(2)}_{32} & W^{(2)}_{33} & W^{(2)}_{34} & W^{(2)}_{35} \end{bmatrix} \begin{Bmatrix} h_{21} \\ h_{22} \\ h_{24} \\ h_{23} \\ h_{25} \end{Bmatrix} + \begin{Bmatrix} b_{21} \\ b_{22} \\ b_{23} \end{Bmatrix}$$

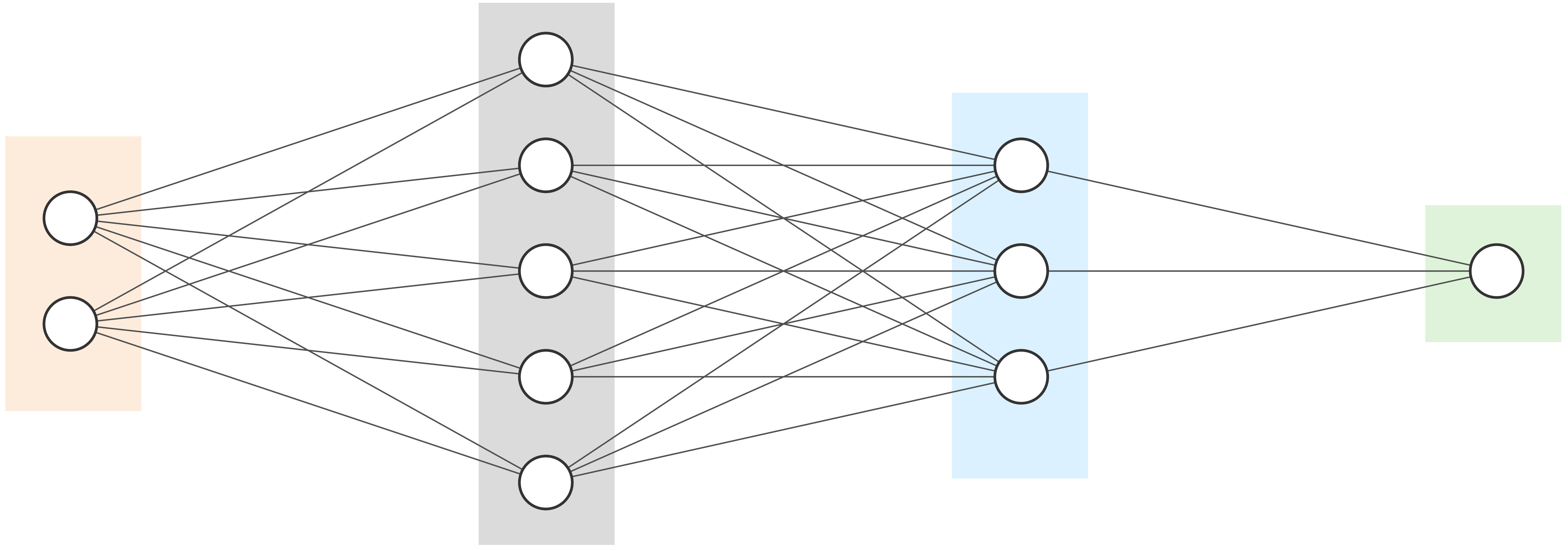
$$\{y\} = \begin{bmatrix} W^{(3)}_{11} & W^{(3)}_{12} & W^{(3)}_{13} \end{bmatrix} \begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix} + \{b_{31}\}$$

Let's see how many parameters are there in the Neural Network below.



$$\underbrace{\begin{Bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{14} \\ h_{15} \end{Bmatrix}}_{5 \times 1} = \underbrace{\begin{bmatrix} W_{11}^{(1)} & W_{12}^{(1)} \\ W_{21}^{(1)} & W_{22}^{(1)} \\ W_{31}^{(1)} & W_{32}^{(1)} \\ W_{41}^{(1)} & W_{42}^{(1)} \\ W_{51}^{(1)} & W_{52}^{(1)} \end{bmatrix}}_{5 \times 2} \underbrace{\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}}_{2 \times 1} + \underbrace{\begin{Bmatrix} b_{11} \\ b_{12} \\ b_{13} \\ b_{14} \\ b_{15} \end{Bmatrix}}_{5 \times 1}$$

Let's see how many parameters are there in the Neural Network below.

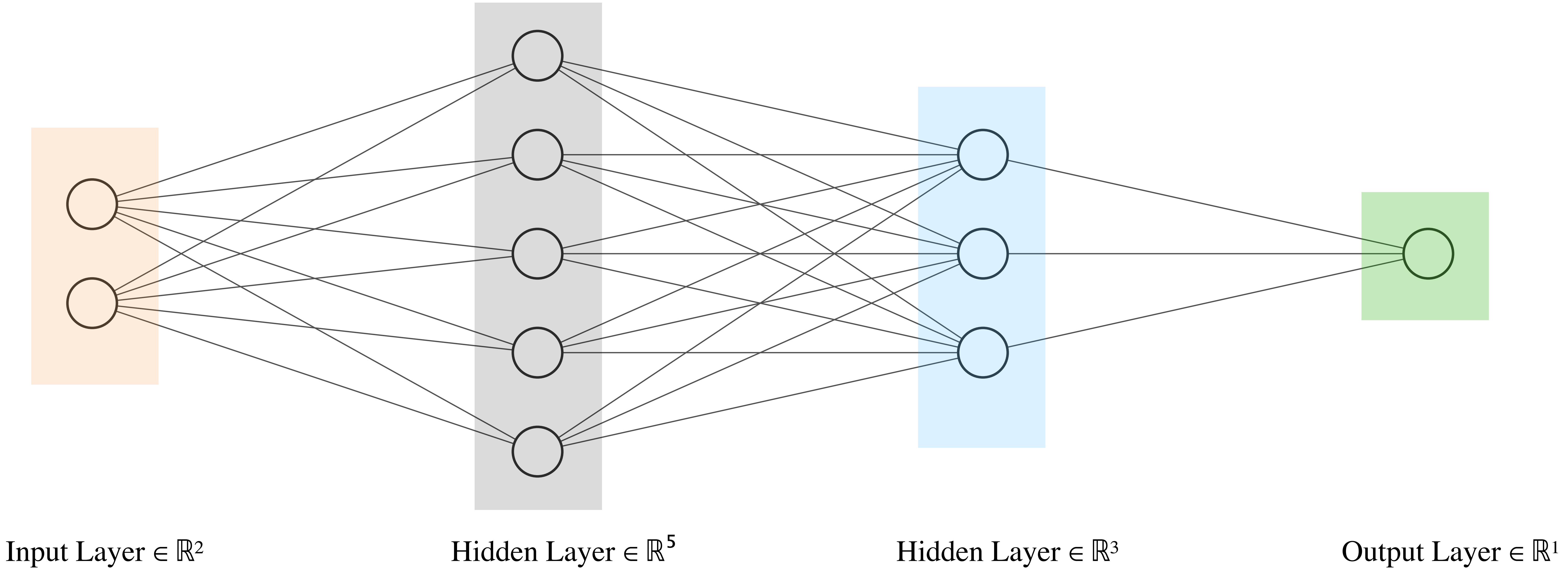


$$\begin{Bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{14} \\ h_{15} \end{Bmatrix} = \begin{bmatrix} W_{11}^{(1)} & W_{12}^{(1)} \\ W_{21}^{(1)} & W_{22}^{(1)} \\ W_{31}^{(1)} & W_{32}^{(1)} \\ W_{41}^{(1)} & W_{42}^{(1)} \\ W_{51}^{(1)} & W_{52}^{(1)} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} + \begin{Bmatrix} b_{11} \\ b_{12} \\ b_{13} \\ b_{14} \\ b_{15} \end{Bmatrix}$$

$$\begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix} = \begin{bmatrix} W_{11}^{(2)} & W_{12}^{(2)} & W_{13}^{(2)} & W_{14}^{(2)} & W_{15}^{(2)} \\ W_{21}^{(2)} & W_{22}^{(2)} & W_{23}^{(2)} & W_{24}^{(2)} & W_{25}^{(2)} \\ W_{31}^{(2)} & W_{32}^{(2)} & W_{33}^{(2)} & W_{34}^{(2)} & W_{35}^{(2)} \end{bmatrix} \begin{Bmatrix} h_{21} \\ h_{22} \\ h_{24} \\ h_{23} \\ h_{25} \end{Bmatrix} + \begin{Bmatrix} b_{21} \\ b_{22} \\ b_{23} \end{Bmatrix}$$

$$\{y\} = \begin{bmatrix} W_{11}^{(3)} & W_{12}^{(3)} & W_{13}^{(3)} \end{bmatrix} \begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix} + \{b_{31}\}$$

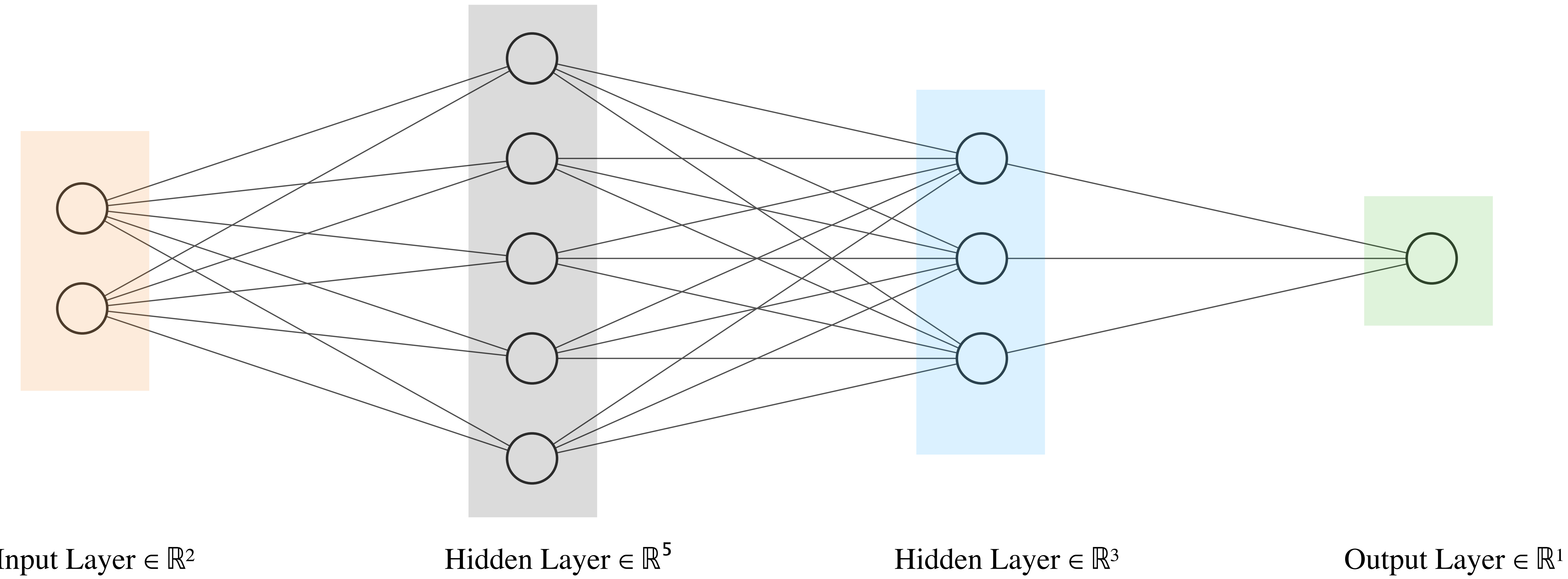
Let's see how many parameters are there in the Neural Network below.



$$\begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix} = \begin{bmatrix} W_{11}^{(2)} & W_{12}^{(2)} & W_{13}^{(2)} & W_{14}^{(2)} & W_{15}^{(2)} \\ W_{21}^{(2)} & W_{22}^{(2)} & W_{23}^{(2)} & W_{24}^{(2)} & W_{25}^{(2)} \\ W_{31}^{(2)} & W_{32}^{(2)} & W_{33}^{(2)} & W_{34}^{(2)} & W_{35}^{(2)} \end{bmatrix} \begin{Bmatrix} h_{21} \\ h_{22} \\ h_{24} \\ h_{23} \\ h_{25} \end{Bmatrix} + \begin{Bmatrix} b_{21} \\ b_{22} \\ b_{23} \end{Bmatrix}$$

3×1 $\quad$ 3×5 $\quad$ 5×1 $\quad$ 3×1

Let’s see how many parameters are there in the Neural Network below.

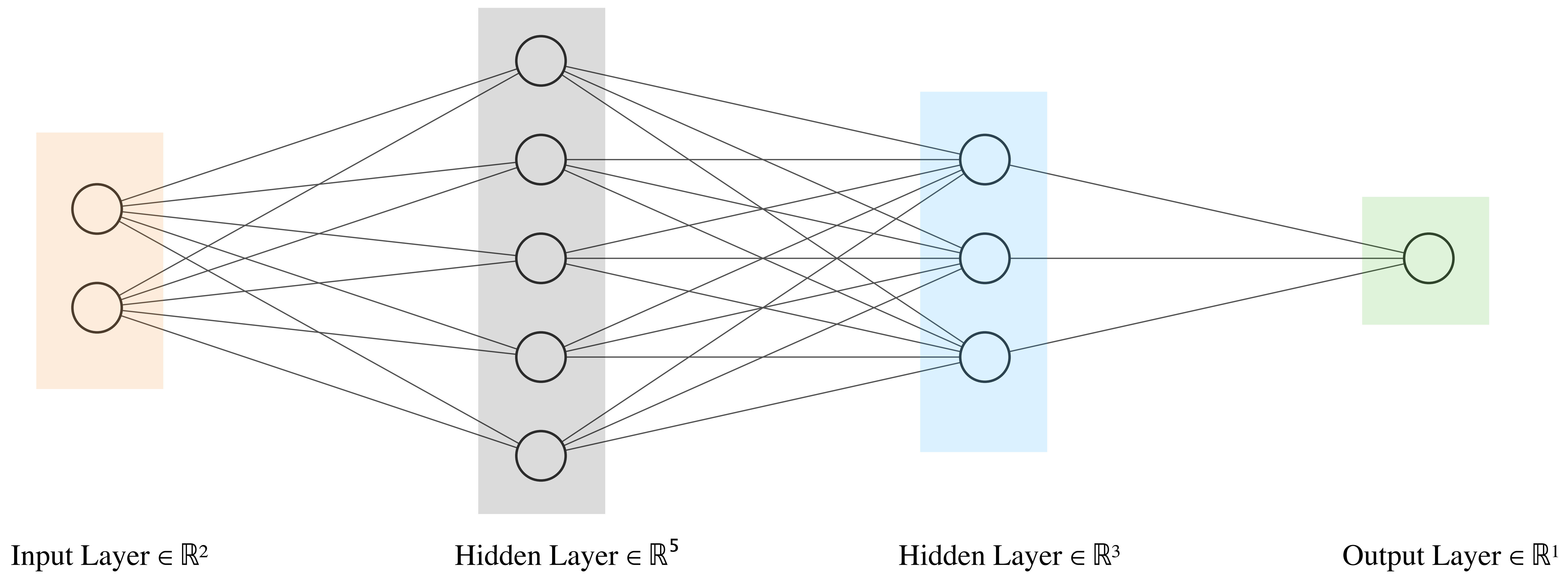


$$\begin{Bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{14} \\ h_{15} \end{Bmatrix} = \begin{bmatrix} W_{11}^{(1)} & W_{12}^{(1)} \\ W_{21}^{(1)} & W_{22}^{(1)} \\ W_{31}^{(1)} & W_{32}^{(1)} \\ W_{41}^{(1)} & W_{42}^{(1)} \\ W_{51}^{(1)} & W_{52}^{(1)} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} + \begin{Bmatrix} b_{11} \\ b_{12} \\ b_{13} \\ b_{14} \\ b_{15} \end{Bmatrix}$$

$$\begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix} = \begin{bmatrix} W_{11}^{(2)} & W_{12}^{(2)} & W_{13}^{(2)} & W_{14}^{(2)} & W_{15}^{(2)} \\ W_{21}^{(2)} & W_{22}^{(2)} & W_{23}^{(2)} & W_{24}^{(2)} & W_{25}^{(2)} \\ W_{31}^{(2)} & W_{32}^{(2)} & W_{33}^{(2)} & W_{34}^{(2)} & W_{35}^{(2)} \end{bmatrix} \begin{Bmatrix} h_{21} \\ h_{22} \\ h_{24} \\ h_{23} \\ h_{25} \end{Bmatrix} + \begin{Bmatrix} b_{21} \\ b_{22} \\ b_{23} \end{Bmatrix}$$

$$\{y\} = \begin{bmatrix} W_{11}^{(3)} & W_{12}^{(3)} & W_{13}^{(3)} \end{bmatrix} \begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix} + \{b_{31}\}$$

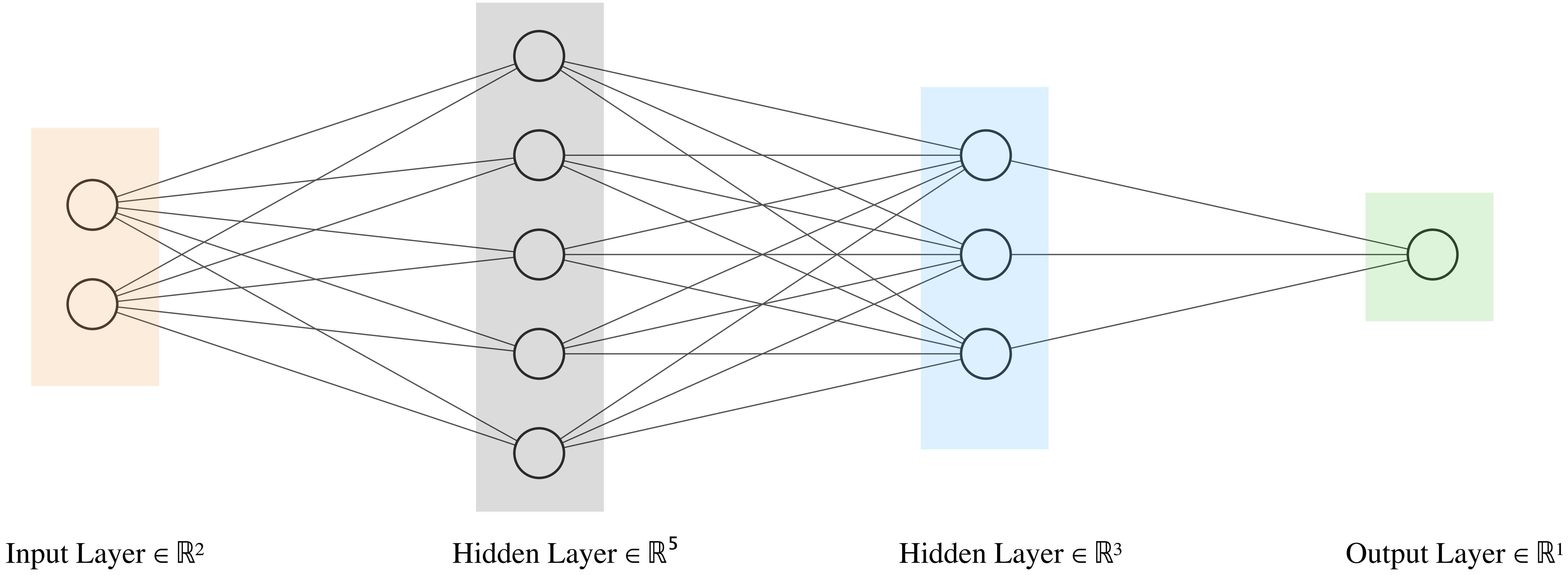
Let’s see how many parameters are there in the Neural Network below.



$$\{y\} = \begin{bmatrix} W_{11}^{(3)} & W_{12}^{(3)} & W_{13}^{(3)} \end{bmatrix} \begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix} + \{b_{31}\}$$

1×1 $\quad 1 \times 3$ $\quad 3 \times 1$ $\quad 1 \times 1$

Let’s see how many parameters are there in the Neural Network below.



$$\underbrace{\begin{Bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{14} \\ h_{15} \end{Bmatrix}}_{5\times 1} = \begin{bmatrix} W^{(1)}_{11} & W^{(1)}_{12} \\ W^{(1)}_{21} & W^{(1)}_{22} \\ W^{(1)}_{31} & W^{(1)}_{32} \\ W^{(1)}_{41} & W^{(1)}_{42} \\ W^{(1)}_{51} & W^{(1)}_{52} \end{bmatrix} \underbrace{\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}}_{2\times 1} + \begin{Bmatrix} b_{11} \\ b_{12} \\ b_{13} \\ b_{14} \\ b_{15} \end{Bmatrix}$$

$$\underbrace{\begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix}}_{3\times 1} = \begin{bmatrix} W^{(2)}_{11} & W^{(2)}_{12} & W^{(2)}_{13} & W^{(2)}_{14} & W^{(2)}_{15} \\ W^{(2)}_{21} & W^{(2)}_{22} & W^{(2)}_{23} & W^{(2)}_{24} & W^{(2)}_{25} \\ W^{(2)}_{31} & W^{(2)}_{32} & W^{(2)}_{33} & W^{(2)}_{34} & W^{(2)}_{35} \end{bmatrix} \underbrace{\begin{Bmatrix} h_{21} \\ h_{22} \\ h_{24} \\ h_{23} \\ h_{25} \end{Bmatrix}}_{5\times 1} + \begin{Bmatrix} b_{21} \\ b_{22} \\ b_{23} \end{Bmatrix}$$

$$\underbrace{\{y\}}_{1\times 1} = \begin{bmatrix} W^{(3)}_{11} & W^{(3)}_{12} & W^{(3)}_{13} \end{bmatrix} \underbrace{\begin{Bmatrix} h_{21} \\ h_{22} \\ h_{23} \end{Bmatrix}}_{3\times 1} + \{b_{31}\}$$

$$5\times 2 \quad 5\times 1 \quad 3\times 5 \quad 3\times 1 \quad 1\times 3 \quad 1\times 1$$

$$10 \quad 5 \quad 15 \quad 3 \quad 3 \quad 1$$

We will use the Lux() package to code up Neural Network in Julia

Input Layer $\in \mathbb{R}^1$

Output Layer $\in \mathbb{R}^1$

```julia
using Lux
model01 = Chain(Dense(1 => 1))
```

```
Chain(
    layer_1 = Dense(1 => 1),                    # 2 parameters
)         # Total: 2 parameters,
          #        plus 0 states.
```

$$NN(x) = Wx + b$$

It is common to initially set W and b to random values.

```julia
rng = Random.default_rng()
parameters, layer_states = Lux.setup(rng, model01)
```

```
((layer_1 = (weight = Float32[-0.34014863;;], bias = Float32[-0.5821972]),), (layer_1 = NamedTuple(),))
```

$$NN(x) = -0.3401x - 0.5822$$

It is common to initially set W and b to random values.

```
rng = Random.default_rng()
parameters, layer_states = Lux.setup(rng, model01)
```

```
((layer_1 = (weight = Float32[-0.34014863;;], bias = Float32[-0.5821972]),), (layer_1 = NamedTuple(),))
```

$$NN(x) = -0.3401x - 0.5822$$

You can explicit specify the value of W and b using the code below

```
parameters.layer_1.weight.=Float32(2.0)
parameters.layer_1.bias.=Float32(1.0)
```

$$NN(x) = 2x + 1$$

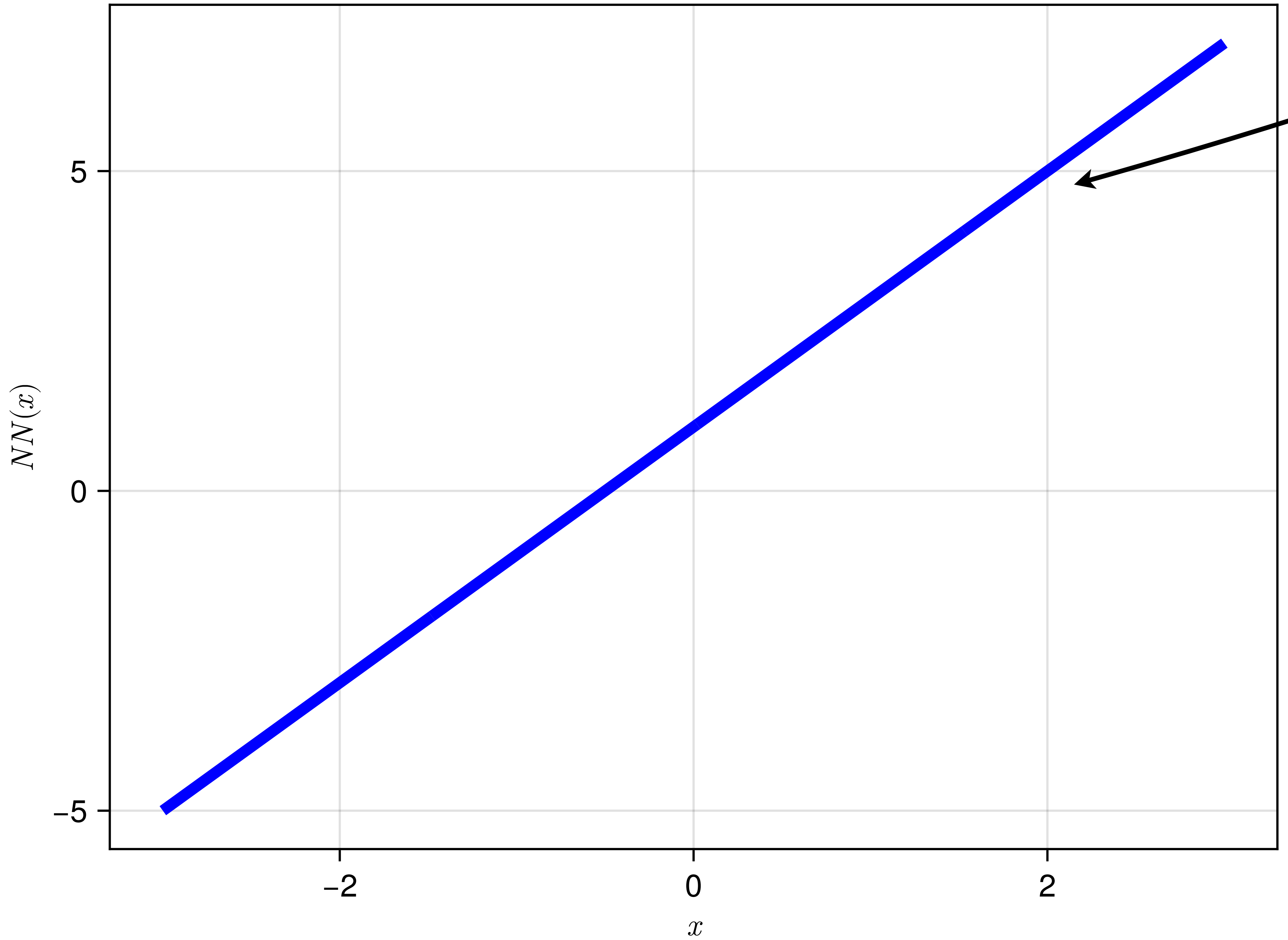
Let's plot the graph of this Neural Network and see how it looks like. Should be a straight line of slope 2 and y-intercept at 1

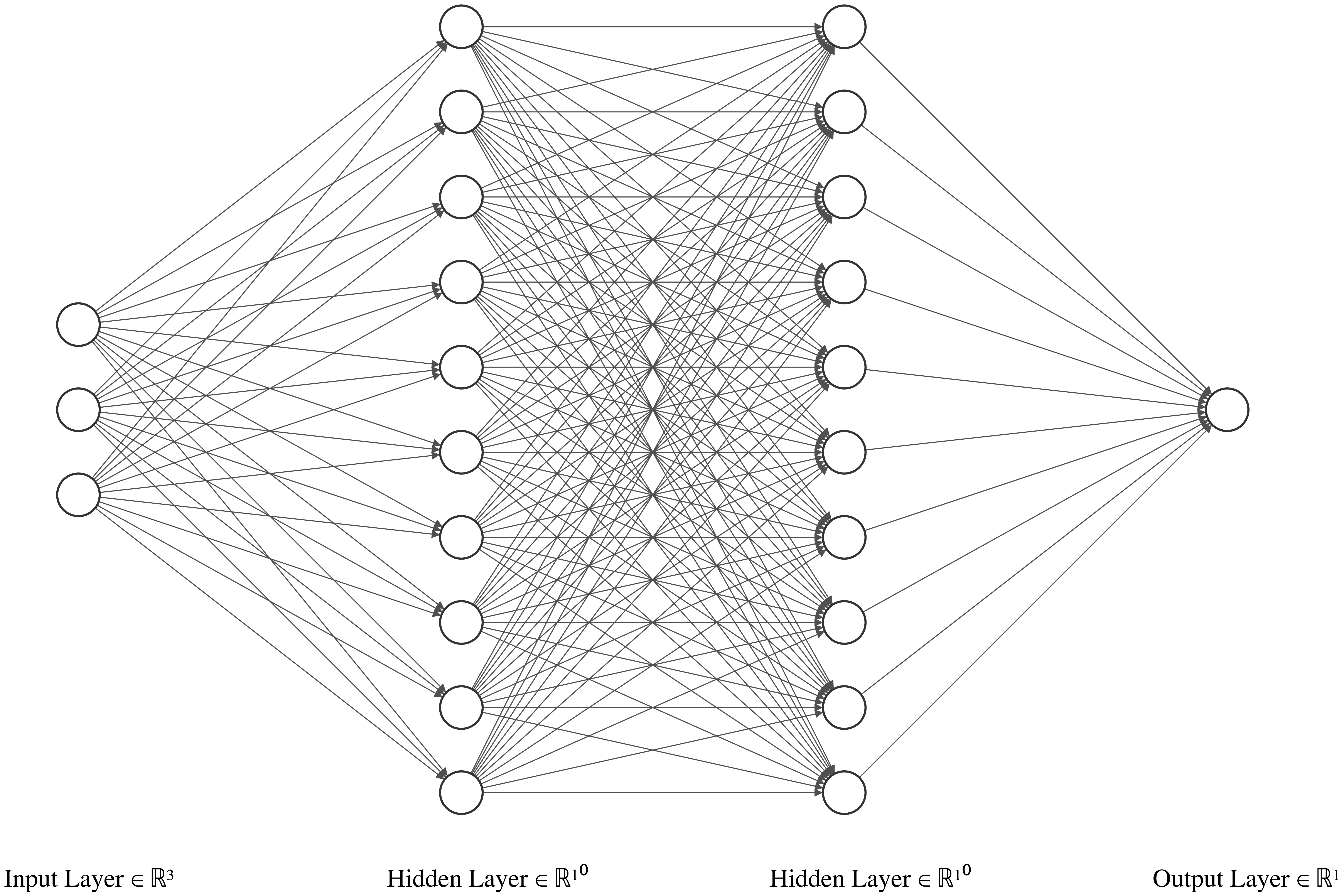
```
parameters.layer_1.weight.=Float32(2.0)
parameters.layer_1.bias.=Float32(1.0)
```

$$NN(x) = 2x + 1$$

Let’s plot the graph of this Neural Network and see how it looks like. Should be a straight line of slope 2 and y-intercept at 1

```
xgrid=-3.0:0.01:3.0
y_prediction,_=model01(xgrid',parameters, layer_states)
fig = Figure()
ax = CairoMakie.Axis(fig[1, 1], xlabel = L"x", ylabel = L"NN(x)")
lines!(ax,xgrid,y_prediction[:];color=:blue,label="prediction",linewidth=5)
fig
```




Input Layer $\in \mathbb{R}^{3}$
Hidden Layer $\in \mathbb{R}^{10}$
Hidden Layer $\in \mathbb{R}^{10}$
Output Layer $\in \mathbb{R}^{1}$